



RIVER VALLEY HIGH SCHOOL

H2 CHEMISTRY

2021 A-Level Suggested Solutions

Paper 1

Question Number	Key
1	B
2	D
3	C
4	B
5	C
6	D
7	B
8	D
9	D
10	A

Question Number	Key
11	C
12	A
13	B
14	B
15	A
16	A
17	C
18	B
19	D
20	C

Question Number	Key
21	A
22	A
23	A
24	B
25	B
26	D
27	B
28	C
29	C
30	D

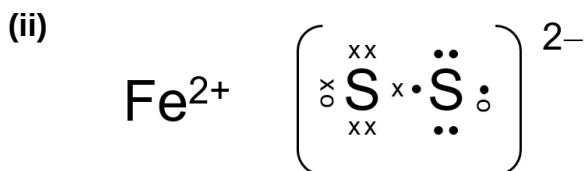
Paper 2

1 (a) (i)
$$A_r \text{ of sulfur} = \left(\frac{63.8}{63.8 + 2.8} \times 31.97 \right) + \left(\frac{2.8}{63.8 + 2.8} \times 33.97 \right)$$

$$= \underline{32.05} \text{ (4 s.f.)}$$

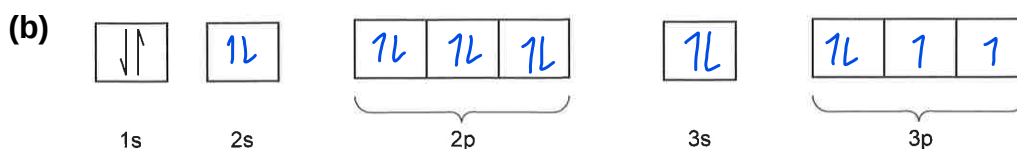
Examiner's Report: Weaker responses were based on all the atoms present in the sample rather than selecting only the sulfur atoms.

[2]



Examiner's Report: Some weaker responses drew covalent structures or did not account for the two extra electrons present in the S₂ anion and showed a double bond between the two sulfur atoms.

[2]



[1]

(c) (i) Enthalpy change when one mole of electrons is removed from one mole of gaseous sulfur atoms to form one mole of gaseous, singly charged sulfur cations.

[2]

(ii) First ionisation energy increases across Period 3. Across the period, nuclear charge increases, while there is negligible increase in shielding effect. (Hence, the effective nuclear charge increases.) The atomic radius also decreases. Therefore, valence electrons are increasingly more attracted by the positive nucleus.

Coulombic repulsion between the paired 3p electrons in sulfur makes it easier to remove one of the paired 3p electrons.

Examiner's Report: Strong responses linked the effect of interelectronic repulsion to the specific location of the electron involved to explain why the ionisation of sulfur does not follow the trend. Weaker responses included statements which did not describe the location of the electron removed or gave an incorrect location or suggested that there was repulsion between the orbitals.

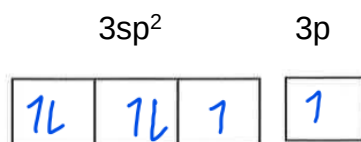
[2]

- (d) (i) The C atom has two sp hybrid orbitals formed via the mixing of its 2s orbital and one of the 2p orbitals. The hybrid orbitals are used to form σ bonds with the S atoms.

The remaining two 2p orbitals remain unhybridised and are used to form π bonds with the S atoms.

[2]

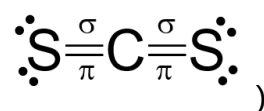
- (ii) Each S atom has three sp^2 hybrid orbitals and one unhybridised 3p orbital.



The hybrid orbital with the unpaired electron is used to form a σ bond with the sp hybrid orbital of C atom.

The unhybridised 3p orbital is used to form a π bond with the unhybridised 2p orbital of C atom.

(This results in the following bonding in CS_2 :



Examiner's Report: Strong answers clearly described the arrangement of the electrons. The use of labelled diagrams proved particularly useful by those finding difficulty conveying understanding in words.

[2]

[Total: 13]

- 2 (a) (i) Down the group, ionic radius increases as the number of principal quantum shells of electrons increases. [1]

- (ii) Cs^+ is much larger than Na^+ and K^+ . Hence, more Cl^- ions can be packed around the larger Cs^+ ion. [1]

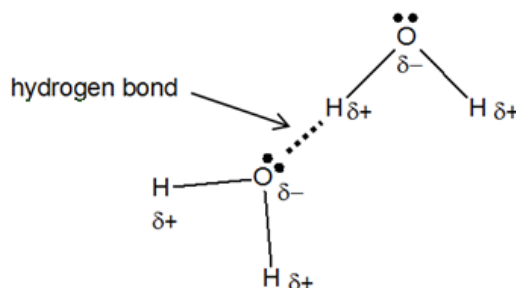
(b)
$$|\text{LE}| \propto \left| \frac{q_+ q_-}{r_+ + r_-} \right|$$

The two factors are:

1. The product of the ionic charges
2. The interionic distance

Factor (1) has a greater effect. Even though the product of the ionic charges of MgO is 4 times that of NaF, and the interionic distance of MgO (0.205 nm) is slightly smaller than that of NaF (0.231 nm), the magnitude of the lattice energy of MgO (3899) is still more than 4 times that of NaF (902). [2]

- (c) (i)



Examiner's Report: Lone pairs on the oxygen atom and the correct partial charges involved in the hydrogen bond were not always shown. Representation of the two water molecules with two hydrogen bonds between the molecules was a common misconception. [2]

- (ii) More energy is released from the formation of favourable ion-dipole interactions between the ions (Na^+ and Cl^-) and polar H_2O molecules than the energy required to overcome the ionic bonds in NaCl(s) and hydrogen bonds between H_2O molecules.

Less energy is released when NaCl is mixed with hexane, as the ions can only form weak instantaneous dipole-induced dipole interactions with non-polar hexane molecules.

Examiner's Report: Detailed answers were seen which demonstrated an excellent understanding of the difference in solubility of sodium chloride in water and in hexane in terms of the energy required to break the relevant forces of attraction, between solute particles and between solvent particles, compared to the amount of energy released by production of solvent-solute interactions. [1]

- (d) (i) There is an increase in disorder as the solid NH_4Cl dissolves to form aqueous NH_4^+ and Cl^- ions. Thus, there are more ways of arranging the particles.

Examiner's Report: Stronger answers focused on the specific process given which related to the resultant formation of aqueous ions. It was more common to see general statements, such as 'increase in disorder' or 'more ways to arrange the particles' with no reference to the process.

[1]

- (ii) $\Delta G^\ominus$ is negative as NH_4Cl is a water-soluble salt. This means that the dissolution process is spontaneous at 298 K.

[1]

- (iii) $\Delta G = \Delta H - T\Delta S$

$$0 = (+15.2) - T\left(+\frac{73.5}{1000}\right)$$

$$T = \underline{207 \text{ K}}$$

At 207 K, H_2O exists as ice, which is unable to dissolve NH_4Cl , rather than liquid water.

[2]

[Total: 11]

3 (a) $E(\text{H-F}) = 562 \text{ kJ mol}^{-1}$

$E(\text{H-Cl}) = 431 \text{ kJ mol}^{-1}$

Since the H-F bond is stronger, it is less likely to dissociate in water. Thus, HF(aq) is a weaker acid than HCl(aq).

[1]

(b)	$\text{HF(aq)} \rightleftharpoons \text{H}^+(\text{aq}) + \text{F}^-(\text{aq})$
initial conc. / mol dm ⁻³	0.0500 0 0
change in conc. / mol dm ⁻³	-x +x +x
eqm conc. / mol dm ⁻³	0.0500 - x x x

$$K_a = \frac{x^2}{0.0500 - x}$$

Since x is small, $0.0500 - x \approx 0.0500$.

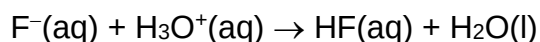
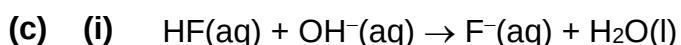
$$5.62 \times 10^{-4} = \frac{x^2}{0.0500}$$

$$x = 5.301 \times 10^{-3}$$

$$\text{pH} = -\lg(5.301 \times 10^{-3})$$

$$= \underline{2.28}$$

[2]



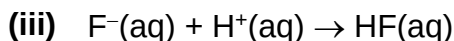
Examiner's Report: Upon the addition of acid some equations expressed were in terms of H^+ ions rather than the H_3O^+ ions specified.

[2]

(ii) Cl^- is a conjugate base from the strong acid HCl, and therefore cannot accept a H^+ . Thus, the mixture is unable to resist a change in pH when a small amount of acid or base is added to it.

Examiner's Report: Many candidates understood that a weak acid, rather than the strong acid, HCl, is required to make a buffer solution. There was confusion in some answers which referred to the role of NaCl. Incorrect statements included 'NaCl is a base' or that 'chloride ions are not a conjugate base of HCl'.

[1]



$$\begin{aligned}\text{Amount of } \text{H}^+ \text{ added} &= \frac{50.0}{1000} \times 0.100 \times 2 \\ &= 0.0100 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Amount of } \text{F}^- \text{ (in } 200 \text{ cm}^3 \text{ of buffer T)} &= \frac{100}{1000} \times 1.78 \\ &= 0.178 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Amount of } \text{F}^- \text{ (in } 75 \text{ cm}^3 \text{ of buffer T)} &= \frac{0.178}{200} \times 75 \\ &= 0.06675 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Resultant amount of } \text{F}^- &= 0.06675 - 0.0100 \\ &= 0.05675 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Resultant } [\text{F}^-] &= \frac{0.05675}{\frac{50.0+75.0}{1000}} \\ &= \underline{0.454 \text{ mol dm}^{-3}}\end{aligned}$$

Examiner's Report: Clear and logical calculations were given by a relatively small number of candidates working from first principles to answer the question. Of those who tackled the question in this way, candidates who performed less well made errors such as either equating the amount of sulfuric acid added to the amount of protons added or using an incorrect final volume of solution to convert the amount of fluoride ions present into concentration. Some weaker responses made inappropriate use of the Henderson-Hasselbalch equation.

[3]

- (d) (i) A smaller pK_a value implies a stronger acid and larger extent of dissociation. Thus, $\text{CCl}_3\text{CO}_2\text{H}$ dissociates to a larger extent than $\text{CH}_3\text{CO}_2\text{H}$.

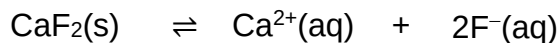
[1]

- (ii) $x = \underline{0.50}$ (any value smaller than 0.66)

$\text{CF}_3\text{CO}_2\text{H}$ is a stronger acid than $\text{CCl}_3\text{CO}_2\text{H}$ as the F atom, being more electronegative and hence more electron-withdrawing, disperses the negative charge on $-\text{COO}^-$ group more effectively than the Cl atom. Hence, CF_3CO_2^- is a more stable anion than $\text{CCl}_3\text{CO}_2^-$.

[1]

- (e) (i) Let the solubility of CaF_2 be $x \text{ mol dm}^{-3}$.



$$\begin{array}{ccc} \text{eqm conc. / mol dm}^{-3} & & \\ & x & 2x \end{array}$$

$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^{-}]^2$$

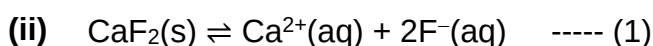
$$3.90 \times 10^{-11} = (x)(2x)^2 = 4x^3$$

$$x = 2.136 \times 10^{-4}$$

$$[\text{F}^{-}] = 2 \times 2.136 \times 10^{-4}$$

$$= \underline{4.27 \times 10^{-4} \text{ mol dm}^{-3}}$$

[2]



In acidic solution, $[\text{H}^{+}]$ increases and the position of equilibrium of (2) shifts right, resulting in a decrease in $[\text{F}^{-}]$. This causes the position of equilibrium of (1) to shift right to replenish some F^{-} . Thus, CaF_2 becomes more soluble.

Examiner's Report: Weaker answers showed some understanding of the process but either lacked all relevant detail or referred to the change to the position of equilibrium with no reference to a relevant equation. Occasionally, answers were given in terms of solute-solvent interactions.

[2]

[Total: 15]

- 4 (a) The C–F and C–C bonds within the polymer are very strong, and require a large amount of energy to overcome.

Examiner's Report: A relatively small number of candidates were able to correctly link the high strength of bonds within the polymer to its unreactive nature. Weaker responses referred to the high strength of C=C bonds which are not present in PTFE or thought that the strength of intermolecular forces was key.

[1]

- (b) $\text{CHClF}_2(\text{g})$ forms radicals which leads to ozone layer depletion.

[1]

- (c) High temperature and low pressure

[1]

- (d) (i) Homolytic fission is a process where the covalent bond is broken such that each of the bonding atoms takes one of the bonding electrons, giving rise to free radicals.

Examiner's Report: Weaker responses made no reference to the breaking of a bond or produced ambiguous statements that suggested that a covalent bond was still present or reformed at the end of the process or that a pair of electrons was transferred to each species.

[1]

- (ii) A: initiation

B: propagation

C: propagation

D: termination

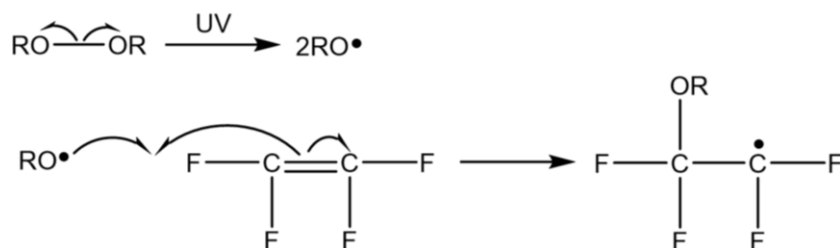
[1]

- (iii) Free radical addition

Examiner's Report: This item proved to be challenging. Correct reference to the free radicals involved in the process was seen in the majority of answers. A small proportion used the information given in the question to deduce that it was an addition reaction rather than substitution reaction.

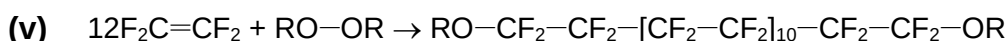
[1]

(iv)



Examiner's Report: The description of the mechanism which represented the homolytic fission of RO—OR was common. The mechanism involved when RO• reacts with the monomer proved more challenging. Weaker responses showed confusion when representing the movement of electrons. These responses were unable to identify the correct radical produced and showed two species at the end of the reaction.

[3]



$$\Delta H_r = \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$$

$$= [(+150) + 12(+610 - 350)] - [2(+360) + 11(+350)]$$

$$= \underline{-1300 \text{ kJ mol}^{-1}}$$

Examiner's Report: This item proved challenging for many. A significant number of candidates selected the correct bond energies involved in the process and used the idea that the overall enthalpy change involved the sum of the bond energies for the bonds broken minus the sum of the bond energies for the bonds formed. Of these only the better performing candidates used correct stoichiometry for this process. Common stoichiometric errors, seen in weaker responses, included breaking only one C=C bond or making an incorrect number of C—C bonds.

[2]

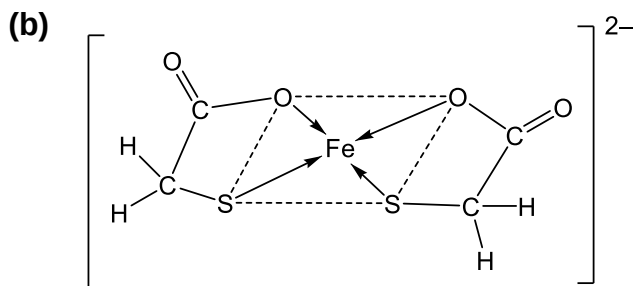
(vi) P would be the major species formed, as it is stabilised by delocalisation of the unpaired electron into the benzene ring.

Examiner's Report: The majority of candidates selected the correct radical; others incorrectly stated that the electron donating nature of the benzene ring was responsible for the difference in stability. Stronger responses appreciated that resonance with the benzene ring was relevant but gave incorrect descriptions of the species involved in the delocalisation process, for example 'a pair of electrons' or 'the radical delocalised into the benzene ring.'

[1]

[Total: 12]

- 5 (a) Thioglycolic acid acts as the reducing agent. The oxidation state of iron decreases from +3 in Fe^{3+} to +2 in Fe^{2+} . [1]



- (c) Ammonia acts as a base to neutralise the acid present and to react with L to form the pink-coloured complex M which can be used to detect excess iron. [1]

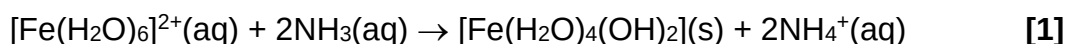
Examiner's Report: Strong responses linked the practical aspects of the method and described the ammonia providing the basic conditions which allowed the excess iron to be detected by forming a pink-coloured product. Many other candidates referred only to ammonia acting as a base and neutralising any acid present. [1]

- (d) $\text{Fe}(\text{SCH}_2\text{CO}_2\text{H})_2 + 2\text{NH}_3 \rightarrow [\text{Fe}(\text{SCH}_2\text{CO}_2)_2]^{2-} + 2\text{NH}_4^+$

Examiner's Report: A significant proportion of candidates described the appropriate species present in the equation. [1]

Weaker responses did not balance the equation given correctly or described inappropriate reactions of L with hydroxide ions rather than with ammonia. [1]

- (e) (i) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{NH}_3(\text{aq}) \rightarrow \text{Fe}(\text{OH})_2(\text{s}) + 4\text{H}_2\text{O}(\text{l}) + 2\text{NH}_4^+(\text{aq})$
Or



- (ii) The green precipitate turns brown due to Fe^{2+} being oxidised to Fe^{3+} by oxygen in the air, forming $\text{Fe}(\text{OH})_3$ which is a brown precipitate. [1]

- (f) Concentration of Fe^{2+} in 250.0 cm³ volumetric flask = 10 x 10.0
= 100 ppm

$$\text{Mass of } \text{Fe}^{2+} \text{ in 250.0 cm}^3 \text{ volumetric flask} = \frac{250}{1000} \times 100$$

$$= 25 \text{ mg}$$

$$\text{Amount of } \text{Fe}^{2+} \text{ in 250.0 cm}^3 \text{ volumetric flask} = \frac{25 \times 10^{-3}}{55.8}$$

$$= 4.480 \times 10^{-4} \text{ mol}$$

[4]

Amount of $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O} = 4.480 \times 10^{-4} \text{ mol}$

$$x = 4.480 \times 10^{-4} \times 392.0$$

$$= \underline{0.176 \text{ g}}$$

Examiner's Report: Weaker responses confused concentration of solutions and amounts of a substance in a specific volume of a solution. Of these, some did not realise that the amount of iron in the 10.00 cm^3 sample is the same as in 100.0 cm^3 of the diluted solution or that the concentration of the 10.00 cm^3 sample is ten times larger than the concentration of the diluted solution. Other candidates used the relative formula mass of the iron salt to find the amount of iron in the diluted sample, which ultimately finds the mass of iron taken from the original solution rather than the mass x .

- (g) Since the shape of the complexes is octahedral, the central iron ion must form 6 bonds with the ligands.

	Y	Fe
<i>mole ratio</i>	$\frac{1}{139.0}$ $= 7.194 \times 10^{-3}$	2.40×10^{-3}
<i>simplest ratio</i>	3	1

Number of bonds formed by Y = 2

	Z	Fe
<i>mole ratio</i>	$\frac{1}{560.0}$ $= 1.786 \times 10^{-3}$	1.78×10^{-3}
<i>simplest ratio</i>	1	1

Number of bonds formed by Z = 6

[2]

[Total: 12]

6 (a) (i)

element	Cr	H	N	S	Cl	O
mass ratio	19.4	5.6	26.1	11.9	13.2	23.8
mole ratio	$\frac{19.4}{52.0}$ = 0.373	$\frac{5.6}{1.0}$ = 5.60	$\frac{26.1}{14.0}$ = 1.86	$\frac{11.9}{32.1}$ = 0.371	$\frac{13.2}{35.3}$ = 0.372	$\frac{23.8}{16.0}$ = 1.49
simplest ratio	$\frac{0.373}{0.371}$ = 1	$\frac{5.60}{0.371}$ = 15	$\frac{1.86}{0.371}$ = 5	$\frac{0.371}{0.371}$ = 1	$\frac{0.372}{0.371}$ = 1	$\frac{1.49}{0.371}$ = 4

Hence, the formula of complex **A** is $\text{CrH}_{15}\text{N}_5\text{SClO}_4$.

[2]

(ii) Shape: octahedral

Bond angle: 90°

[1]

(iii) $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$

[1]

(iv) $[\text{Cr}(\text{NH}_3)_5\text{SO}_4]^+$

Examiner's Report: This question proved to be challenging. Only the stronger responses realised the significance of the test with silver ions to deduce that the chloride ions cannot be present in the cation. Of these, some identified the correct species present in the cation, which is not required, with the incorrect overall charge.

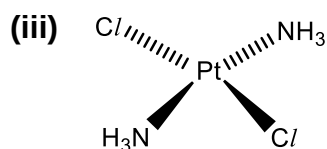
[1]

(b) (i) The high concentration of Cl^- in blood will shift the position of equilibrium in equation 6.1 to the left, hence very little ligand exchange occurs in the blood.

[1]

(ii) Each of the 2 ligands has at least one lone pair of electrons to form only one co-ordinate bond with the Pt^{2+} ion.

[1]



[1]

(iv) In transplatin, after the chloride ions are exchanged for water molecules, the water ligands are on opposite sides of the Pt^{2+} ion/ at different positions/ are further away from each other. Therefore they cannot be substituted with the two specific ligands at the same specific distance on the DNA molecule.

[1]

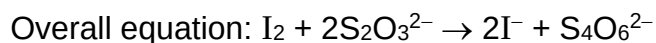
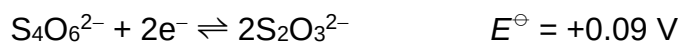
- (c) **D** contains an internal plane of symmetry and has no enantiomerism. Therefore, the solution before heating does not rotate plane-polarised light.

C and **E** are a pair of enantiomers. Since the solution after heating contains an equal amount of both enantiomers, it is a racemic mixture and therefore does not rotate plane-polarised light too.

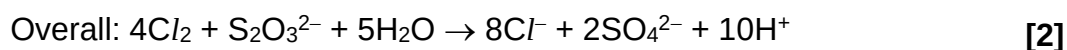
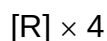
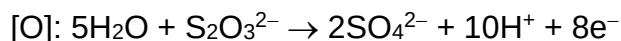
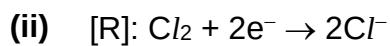
[3]

[Total: 12]

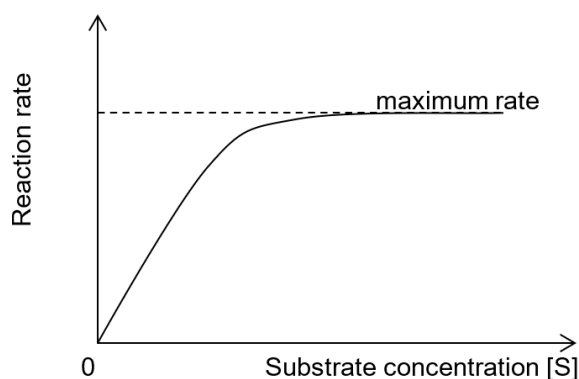
Paper 3 Section A



$E^\ominus_{\text{cell}} = (+0.54) - (+0.09) = \underline{+0.45 \text{ V}}$ [1]



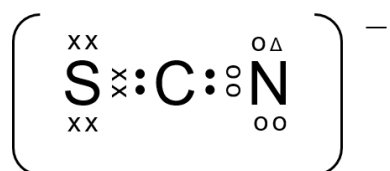
(b)



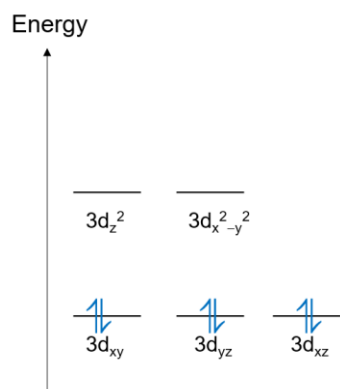
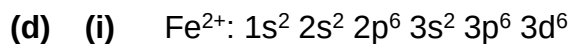
At low [S], the reaction is first-order with respect to the substrate because the rate of enzyme-substrate complex formation is directly proportional to [S].

At high [S], adding more substrate cannot accelerate the reaction as all the active sites on the enzyme molecules are saturated. The rate of reaction then becomes independent of [S] and the reaction is zero-order with respect to the substrate. [2]

(c)



[1]



[1]

- (ii) The H_2O ligand in deoxyhaemoglobin and O_2 ligand in oxyhaemoglobin cause the 3d orbitals to split into two sets of non-degenerate orbitals with different energies. The small energy gap (ΔE) for deoxyhaemoglobin and that for oxyhaemoglobin are different.

Hence, different wavelengths in the visible spectrum are absorbed by deoxyhaemoglobin and oxyhaemoglobin when an electron from a lower energy d-orbital is promoted to a higher energy empty d-orbital during d-d transition. The colours observed are the complementary of the colours absorbed.

[2]

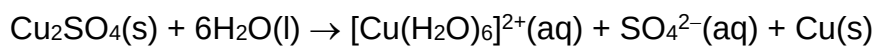
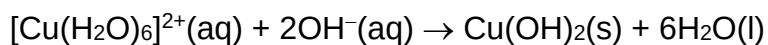
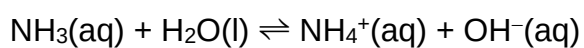
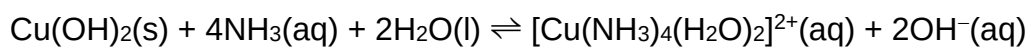
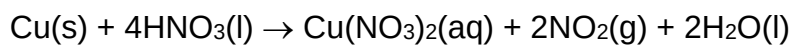
- (iii) When 50% of Hb is converted into $\text{Hb}(\text{O}_2)_4$, $[\text{Hb}] = [\text{Hb}(\text{O}_2)_4]$.

$$\begin{aligned}
 K_c &= \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}][\text{O}_2]^4} \\
 &= \frac{1}{(7.82 \times 10^{-3})^4} \\
 &= \underline{2.67 \times 10^8 \text{ mol}^{-4} \text{ dm}^3}
 \end{aligned}$$

[1]

(e)

G	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$
H	Cu
I	$\text{Cu}(\text{OH})_2$
J	$[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$
K	$\text{Cu}(\text{NO}_3)_2$

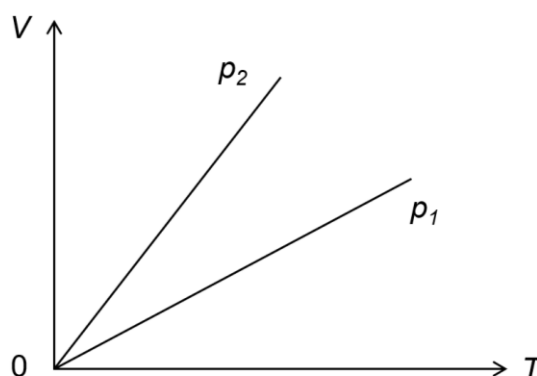
**(G)****(H)****(G)****(I)****(I)****(J)****(H)****(K)****[5]****[Total: 15]**

- 2 (a) (i) 1. The gas particles have negligible volume compared to the volume of the container.
 2. The gas particles exert negligible intermolecular forces on one another.
 3. Collisions between gas particles are perfectly elastic. [2]

(ii) $pV = nRT$

$$V = \left(\frac{nR}{p}\right)T$$

Since n and R are constants, the expected graph is a upward sloping straight line passing through the origin. When p is larger, the straight line has a gentler gradient.



Examiner's Report: A common error was a graph for real gases deviating from ideal behaviour. [2]

(b) $pV = nRT = \frac{mRT}{M}$

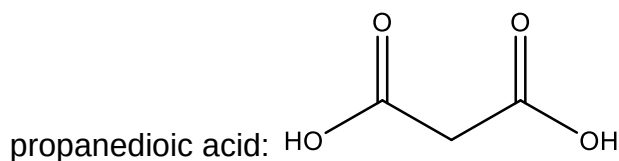
$$\rho = \frac{m}{V} = \frac{pM}{RT} = \frac{(700)(12.0 + 16.0 + 16.0)}{(8.31)(-65 + 273)}$$

$$= 17.8 \text{ g m}^{-3}$$

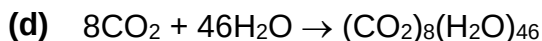
$$= \underline{1.78 \times 10^{-5} \text{ g cm}^{-3}}$$

Examiner's Report: Common errors included calculating the density in g m^{-3} . [2]

(c)



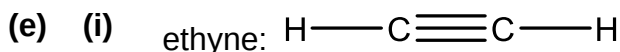
[2]



$$\begin{aligned}\text{Amount of } (\text{CO}_2)_8(\text{H}_2\text{O})_{46} &= \frac{650 \times 10^3}{8(12.0 + 16.0 + 16.0) + 46(1.0 + 1.0 + 16.0)} \\ &= 550.8 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Volume of CO}_2 &= 550.8 \times 8 \times 24 \\ &= \underline{1.06 \times 10^5 \text{ dm}^3}\end{aligned}$$

[2]



Ethyne is a linear molecule and both C atoms in ethyne are sp hybridised.

One sp orbital of the C atom overlaps head-on with the sp orbital of the neighbouring C atom, forming a C—C σ bond. The remaining sp orbital of the C atom overlaps head-on with the 1s orbital of a H atom, forming a C—H σ bond.

The two unhybridised 2p orbitals of the C atom overlap sideways with the two unhybridised 2p orbitals of the neighbouring C atom, forming two C—C π bonds.

[3]

(ii)	$2\text{CH}_4(\text{g})$	$\rightleftharpoons$	$\text{C}_2\text{H}_2(\text{g})$	+	$3\text{H}_2(\text{g})$
initial amount / mol	0.800		0		0
change in amount / mol	$-0.6(0.800)$ $= -0.48$		+0.24		+0.72
eqm amount / mol	0.32		0.24		0.72

$$\begin{aligned}\text{Total amount of gas at eqm} &= 0.32 + 0.24 + 0.72 \\ &= 1.28 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Partial pressure of CH}_4 &= \frac{0.32}{1.28} \times 2.50 \times 10^6 \\ &= 6.25 \times 10^5 \text{ Pa}\end{aligned}$$

$$\begin{aligned}\text{Partial pressure of C}_2\text{H}_2 &= \frac{0.24}{1.28} \times 2.50 \times 10^6 \\ &= 4.6875 \times 10^5 \text{ Pa}\end{aligned}$$

$$\begin{aligned}\text{Partial pressure of H}_2 &= \frac{0.72}{1.28} \times 2.50 \times 10^6 \\ &= 1.40625 \times 10^6 \text{ Pa}\end{aligned}$$

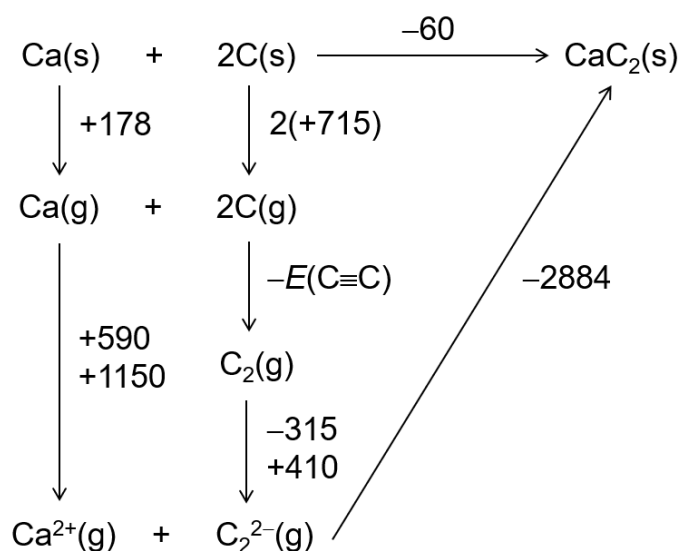
[4]

$$K_p = \frac{(4.6875 \times 10^5)(1.40625 \times 10^6)^3}{(6.25 \times 10^5)^2}$$

$$= \underline{3.34 \times 10^{12} \text{ Pa}^2}$$

Examiner's Report: Most candidates correctly calculated the number of moles of the species at equilibrium, constructed the expression for K_p and deduced the units. However, some did not calculate mole fractions or partial pressures. A common error was in not using mole fractions in the calculation.

(f)



$$-60 = +178 + 590 + 1150 + 2(+715) - E(\text{C}\equiv\text{C}) - 315 + 410 - 2884$$

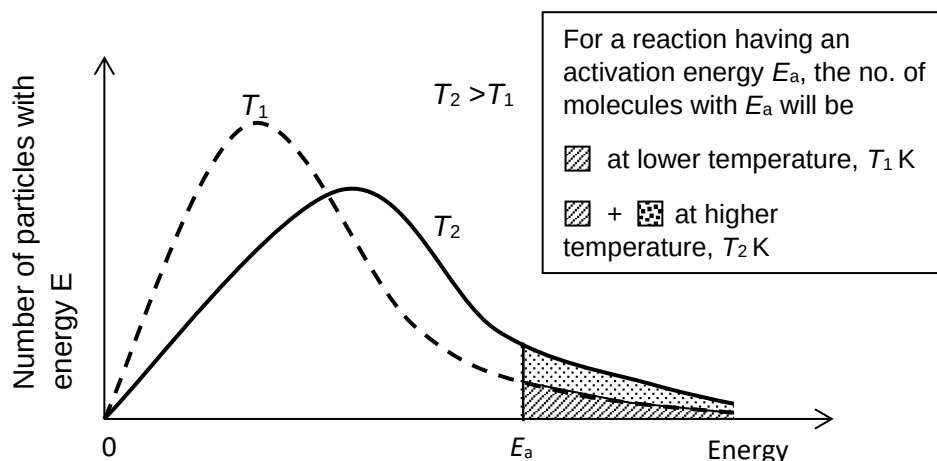
$$E(\text{C}\equiv\text{C}) = \underline{+619 \text{ kJ mol}^{-1}}$$

Examiner's Report: Many candidates found this challenging. Most candidates were able to extract the ionisation energies for calcium and use the correct multiplier for the enthalpy change of sublimation of C(s). Many sketched out a Born Haber cycle but did not include a calculation. Common errors were the incorrect sign usage for the values.

[4]

[Total: 21]

3 (a)



An increase in temperature increases the average kinetic energy of the reacting particles. Hence, the number of particles with energy equal to or greater than the activation energy increases significantly.

This results in an increase in the frequency of effective collisions between the reactant particles. Hence, the rate of reaction and rate constant increases with an increase in temperature.

[3]

(b) Assuming the experimentally-determined rate equation is given by

$$\text{rate} = k [A]^m [B]^n$$

the order of the reaction is $(m + n)$ i.e. sum of the powers of the reactants' concentration term in the rate equation and the rate constant is represented by k i.e. the proportionality constant in the experimentally determined rate equation.

Examiner's Report: Many candidates were not precise enough in their definitions. Errors included omitting key words in their definitions such as 'concentration', 'rate equation' or 'proportionality'. Order of reaction was better known than rate constant.

[2]

(c) (i) Step 1: $\text{Rate} = k_f[\text{O}_3]$

Step 2: $\text{Rate} = k_2[\text{O}_3][\text{O}]$

[1]

(ii) $k_f[\text{O}_3] = k_r[\text{O}][\text{O}_2]$

$$[\text{O}] = \frac{k_f [\text{O}_3]}{k_r [\text{O}_2]}$$

[1]

(iii) $\text{Rate} = k_2[\text{O}_3][\text{O}]$

$$= k_2[\text{O}_3] \frac{k_f [\text{O}_3]}{k_r [\text{O}_2]}$$

$$= k' \frac{[\text{O}_3]^2}{[\text{O}_2]} \text{ where } k' = \frac{k_2 k_f}{k_r}$$

Order of reaction with respect to $[\text{O}_3] = 2$

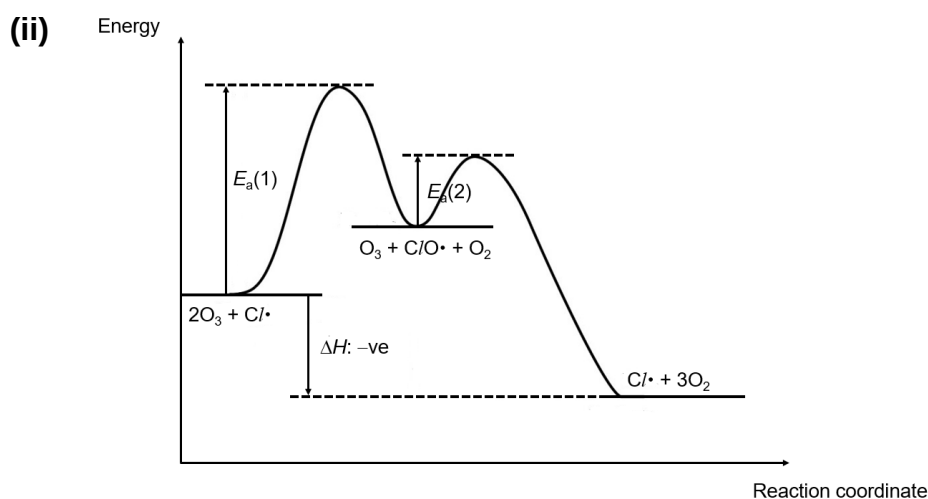
Order of reaction with respect to $[\text{O}_2] = -1$

Overall order of reaction = 1

[2]

- (d) (i) $\text{Cl}\cdot$ is a catalyst as it is consumed in step 1 and regenerated in step 2.

[1]



Examiner's Report: A common error was omitting the intermediate in their energy profile diagram.

[2]

- (e) (i) Adsorption: reactant particles form weak bonds/ interactions to the active sites on the catalyst surface. This weakens or breaks the covalent bonds within the reactant molecules, making them more reactive.

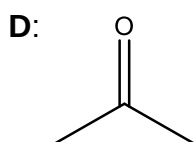
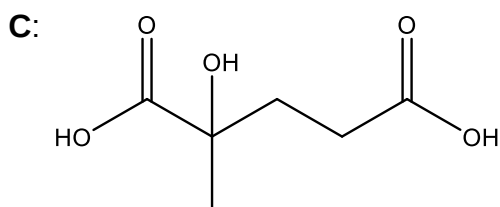
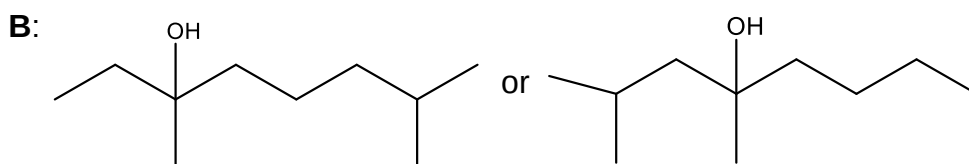
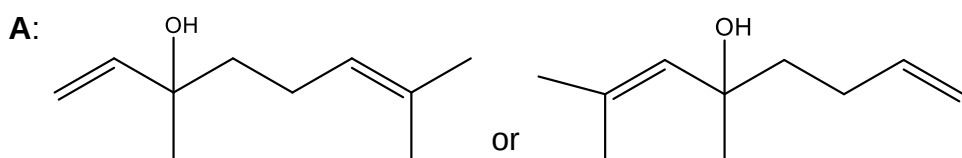
Reaction: the reactant particles are held on the catalyst surface in close proximity and in the correct orientation so that they can readily react.

Desorption: the weak bonds/ interactions between product particles and catalyst surface are broken. The products diffuse away from the surface of the catalyst and the active sites become available again.

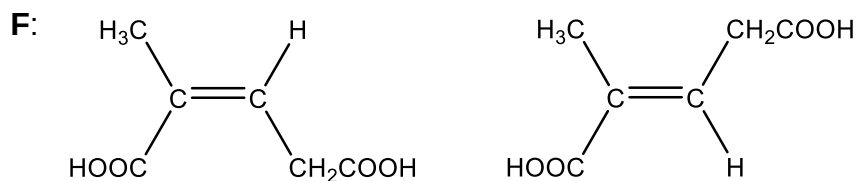
Examiner's Report: A common error was stating that the intermolecular forces between the molecules, instead of the covalent bonds, were weakened as a result of adsorption.

[2]

- (ii) • **A** undergoes reduction with excess hydrogen and Ni to form **B** and **A** undergoes oxidation with ozone, cleaving C=C to give three organic compounds
 ⇒ **A** contains 2 C=C bonds
- **C** undergoes elimination with hot concentrated H₂SO₄ to form **F**, lost a chiral centre after the reaction.
 ⇒ **C** contains —OH group bonded to the chiral centre
 ⇒ **A** contains —OH group
 ⇒ **F** is an alkene which exhibits cis-trans isomerism
- **C** does not undergo oxidation with hot acidified potassium dichromate(VI)
 ⇒ **C** is a tertiary alcohol
- **D** undergoes oxidation with alkaline aqueous iodine but does not undergo oxidation with Fehling's reagent
 ⇒ **D** is a ketone which contains —COCH₃ group
- **C, E and F** undergo acid-base reaction with aqueous Na₂CO₃
 ⇒ **C, E and F** are carboxylic acids



E: HCOOH



Examiner's Report: Many candidates found this challenging. Most gave good explanations, clearly linking the information to the possible functional group present. However, it was rare to see both stereoisomers of **F**.

The types of reactions were generally well known, although common errors included hydrolysis (**C**→**F**), acid-carbonate (**C/E/F** with Na_2CO_3) or omitting oxidation with the iodoform reaction.

Candidates found deducing the structures of **A** to **F** more challenging. Structures **D** and **E** were more frequently known; **A** and **B** were less well known.

[Total: 24]

Paper 3 Section B

4 (a) For halogen X,

Thermal decomposition reaction: $2\text{HX} \rightarrow \text{H}_2 + \text{X}_2$

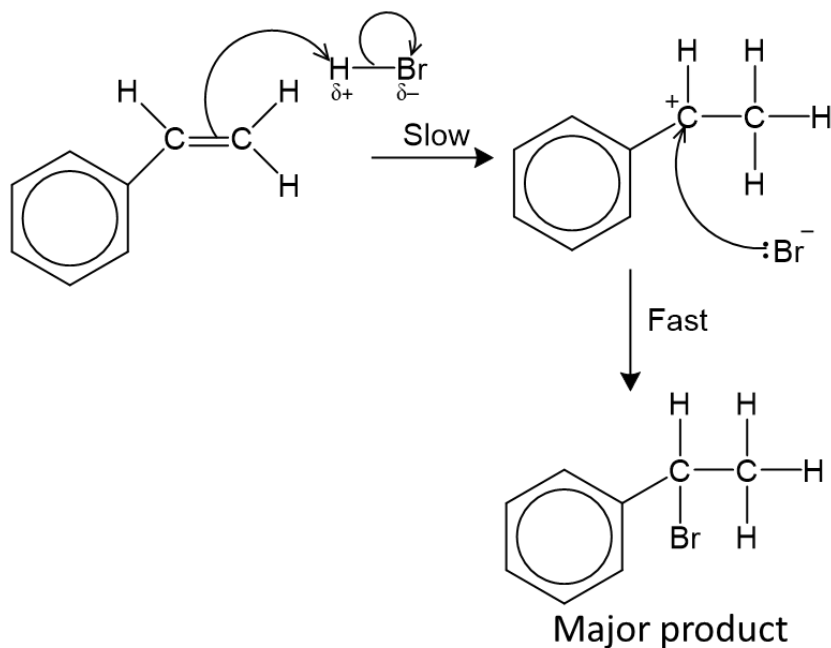
Thermal stability decreases from HCl to HBr to HI .

The thermal stability of hydrogen halides decreases down the Group due to decreasing H–X bond energy.

The size of the halogen atoms increases from Cl to I and their valence orbitals become more diffused. This results in less effective overlap of the orbitals between the small H atom and the larger halogen atom and less energy is required to break the weaker H–X bond.

[3]

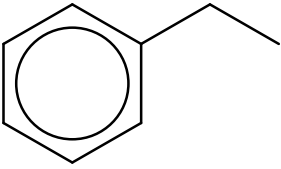
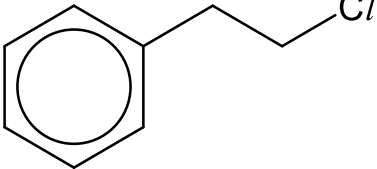
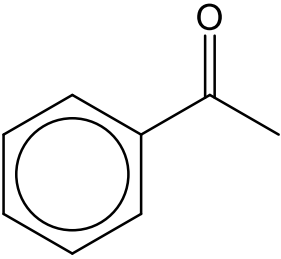
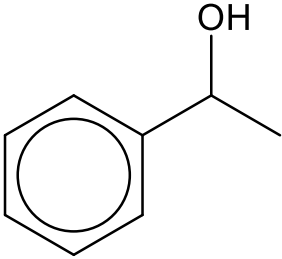
(b) Electrophilic addition mechanism:



A benzylic cation is more stable as the positive charge on the carbocation can delocalise into the benzene ring.

[3]

(c) (i)

 Note: There are multiple correct structures for Q and R depending on the chosen reagents for step 1 and 2.	
Q	R
	
S	T

[4]

(ii) Step 1: $\text{CH}_3\text{CH}_2\text{Cl}$, anhydrous AlCl_3 Step 2: Limited $\text{Cl}_2(\text{g})$, UV lightStep II: LiAlH_4 in dry ether, r.t.p.Step III: Excess concentrated H_2SO_4 , heat

[4]

(d) Isomer 3-position would be formed the least.

Isomer 4-position would be formed the most.

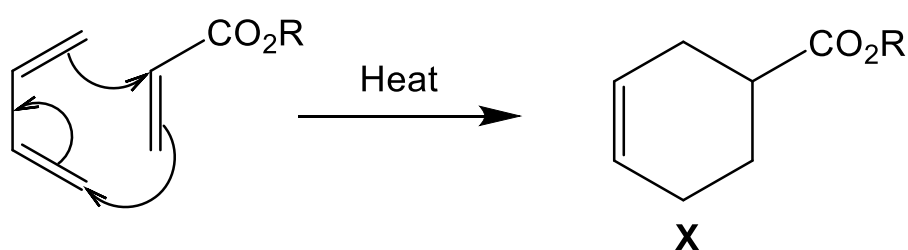
 $(\text{CH}_3)_3\text{C}-$ group is electron donating as well as bulky.

Isomer 4-position is the most favoured as the carbocation intermediate is stabilised by the electron donating effect of $(\text{CH}_3)_3\text{C}-$ group, while the position of substitution is the furthest away, presenting the least amount of steric hindrance.

Isomer 3-position is not favoured as the carbocation intermediate is not stabilised via a direct bond with the $(\text{CH}_3)_3\text{C}-$ group.

[3]

(e) (i)

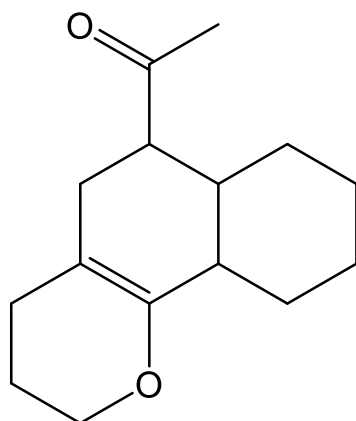


[1]

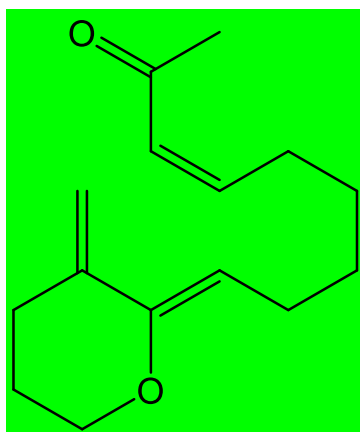
- (ii) Compound **X** has a chiral carbon and is produced as a racemic mixture (as the substituted alkene can attack the diene from above and below the planar diene).

[1]

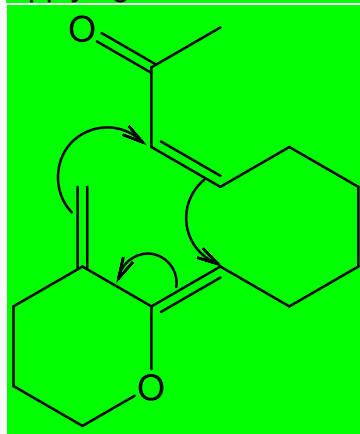
(iii)



Thought process:

Rearranging, compound **W** would look like this:

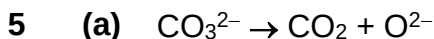
Applying Diels-Alder reaction:



which would lead to the stated answer.

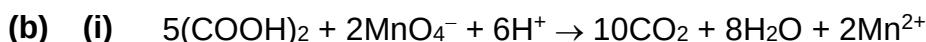
[1]

[Total: 20]

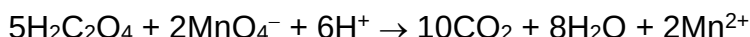


Down the Group, the radius of the metal cation, M^{2+} , increases and its charge density decreases. As a result, the ability of M^{2+} to polarise the electron cloud of the large CO_3^{2-} anion decreases and the C–O bonds are weakened to a smaller extent. Hence thermal stability of the Group 2 carbonates increases down the Group.

Examiner's Report: Candidates performed well on this question. They were able to recall and explain the trend of thermal stability clearly. Some candidates found the equation for the decomposition of the carbonate difficult.



OR



(ii) Amount of MnO_4^- used = $\frac{22.4}{1000} \times 0.00100$

$$= 2.24 \times 10^{-5} \text{ mol}$$

$$\text{Amount of } \text{H}_2\text{C}_2\text{O}_4 \text{ present} = \frac{5}{2} \times 2.24 \times 10^{-5}$$

$$= 5.60 \times 10^{-5} \text{ mol}$$



Note: equation is given in (iii)

$$\text{Amount of } \text{Ca}^{2+} \text{ present} = 5.60 \times 10^{-5} \text{ mol}$$

$$\text{Mass of } \text{Ca}^{2+} \text{ present} = 5.60 \times 10^{-5} \times 40.1$$

$$= 2.2456 \times 10^{-3} \text{ g}$$

$$= \underline{2.25 \text{ mg}} \text{ (to 3 s.f.)}$$

(iii) $K_{\text{sp}} = 10^{-\left(\frac{\Delta G}{2.3RT}\right)}$

$$= 10^{-\left(\frac{+49.2 \times 1000}{2.3 \times 8.31 \times (25 + 273)}\right)}$$

$$= \underline{2.30 \times 10^{-9} \text{ (mol}^2 \text{ dm}^{-6}\text{)}}$$

Examiner's Report: This question was found to be difficult by most candidates. A common incorrect answer was 0.980 (no conversion of ΔG into J mol^{-1}).

(c) (i)
$$\text{pH} = \text{p}K_{\text{a}} + \lg \frac{[\text{lactate ion}]}{[\text{lactic acid}]}$$

$$6.50 = -\lg(1.38 \times 10^{-4}) + \lg \frac{(0.028)}{[\text{lactic acid}]}$$

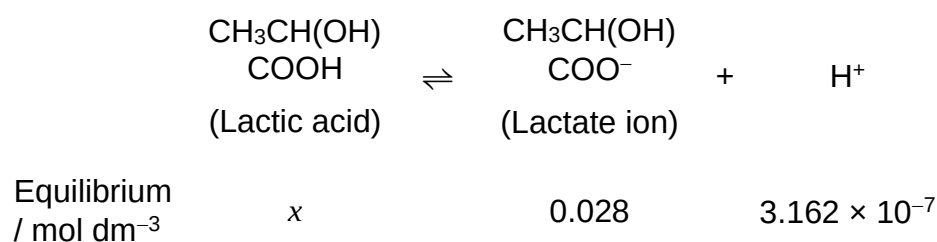
$$[\text{lactic acid}] = 6.416 \times 10^{-5}$$

$$= 6.42 \times 10^{-5} \text{ mol dm}^{-3} \text{ (to 3 s.f.)}$$

OR

$$\text{pH} = 6.50$$

$$[\text{H}^+] = 10^{-6.50} = 3.162 \times 10^{-7} \text{ mol dm}^{-3}$$



$$K_{\text{a}} = \frac{[\text{Lactate ion}][\text{H}^+]}{[\text{Lactic acid}]}$$

$$1.38 \times 10^{-4} = \frac{(0.028)(3.162 \times 10^{-7})}{x}$$

$$[\text{Lactic acid}] = x$$

$$= 6.416 \times 10^{-5}$$

$$= \underline{6.42 \times 10^{-5} \text{ mol dm}^{-3}} \text{ (to 3 s.f.)}$$

Examiner's Report: This question was usually answered well. Many candidates were able to calculate $[\text{H}^+]$ from the given pH. However, fewer were able to use their answer with an expression for K_{a} (or the Henderson Hasselbalch equation) to calculate $[\text{lactic acid}]$.

(ii) Lactic acid + urea $\rightleftharpoons$ lactate ion + urea-H⁺

$$K_c = \frac{[\text{Lactate ion}][\text{urea-H}^+]}{[\text{Lactic acid}][\text{urea}]}$$

Lactic acid + H₂O $\rightleftharpoons$ lactate ion + H₃O⁺

$$K_a = \frac{[\text{Lactate ion}][\text{H}_3\text{O}^+]}{[\text{Lactic acid}]}$$

Urea + H₂O $\rightleftharpoons$ urea-H⁺ + OH⁻

$$K_b = \frac{[\text{urea-H}^+][\text{OH}^-]}{[\text{urea}]}$$

$$K_a \times K_b = \frac{[\text{lactate ion}][\text{H}_3\text{O}^+]}{[\text{lactic acid}]} \times \frac{[\text{urea-H}^+][\text{OH}^-]}{[\text{urea}]}$$

$$K_a \times K_b = \frac{[\text{lactate ion}][\text{urea-H}^+]}{[\text{lactic acid}][\text{urea}]} \times [\text{H}_3\text{O}^+][\text{OH}^-]$$

$$K_a \times K_b = K_c \times K_w$$

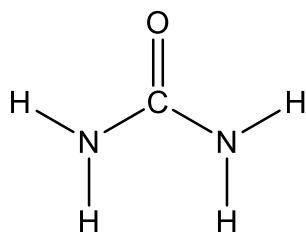
$$K_b = \frac{K_c \times K_w}{K_a}$$

$$K_b = \frac{(1.74 \times 10^{-4})(1.00 \times 10^{-14})}{1.38 \times 10^{-4}}$$

$$K_b = 1.26 \times 10^{-14} \text{ mol dm}^{-3} \text{ (to 3 s.f.)}$$

Examiner's Report: Many candidates found this question challenging. Most answers were simply multiples of K_a and K_b which did not involve K_w .

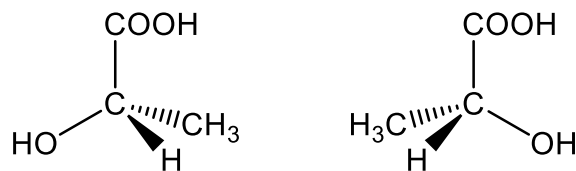
(iii)



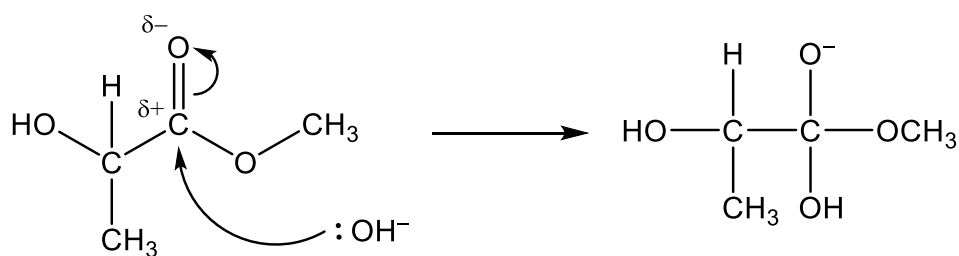
Number of σ bonds: 7

Number of π bonds: 1

(d) (i)

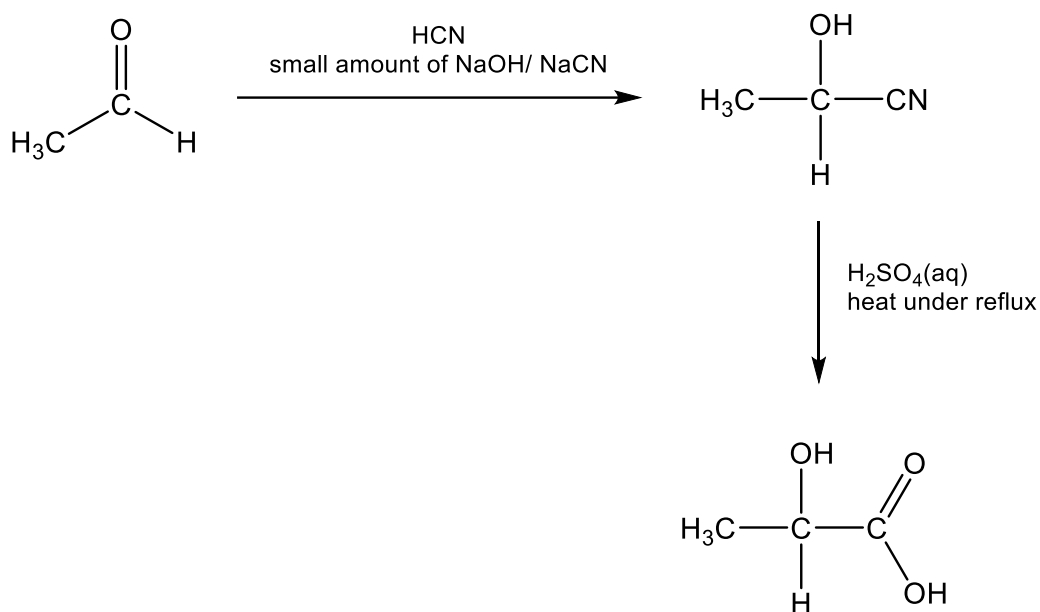


(ii)

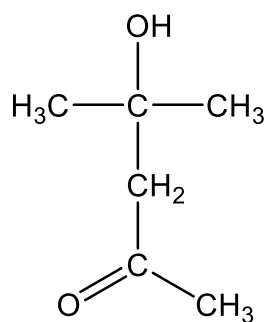


Examiner's Report: Common errors were omitting the dipole in the C=O bond and breaking the ester C–O bond instead of the C=O bond.

(e)

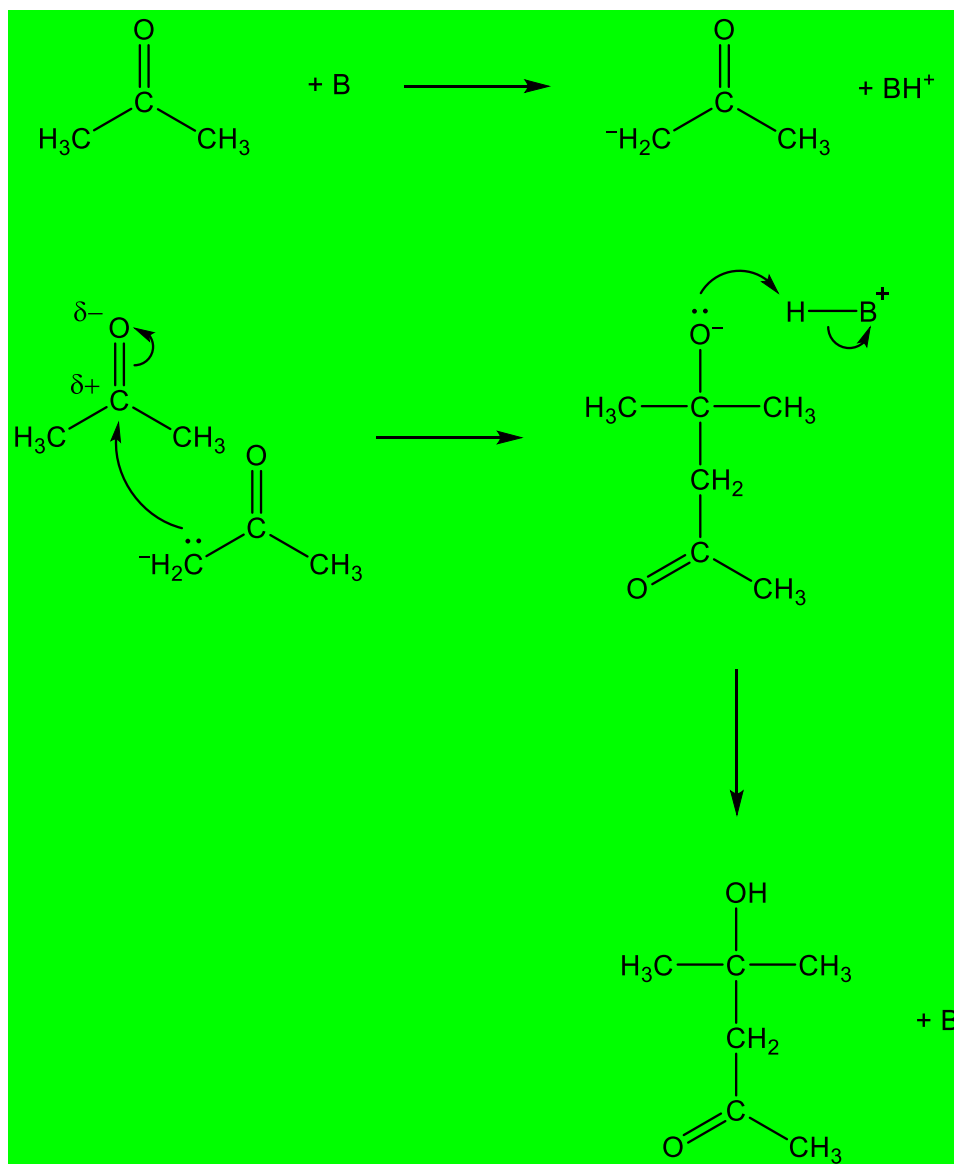


(f) (i)

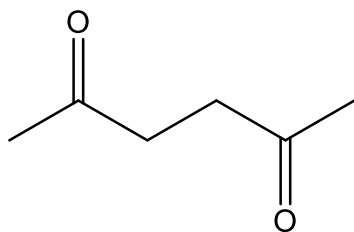


Examiner's Report: Many candidates found this question challenging. Some used propanal instead of propanone.

Teachers' Note: The mechanism is not required but included here for ease of discussion and to bring students' awareness that this is a nucleophilic addition mechanism.

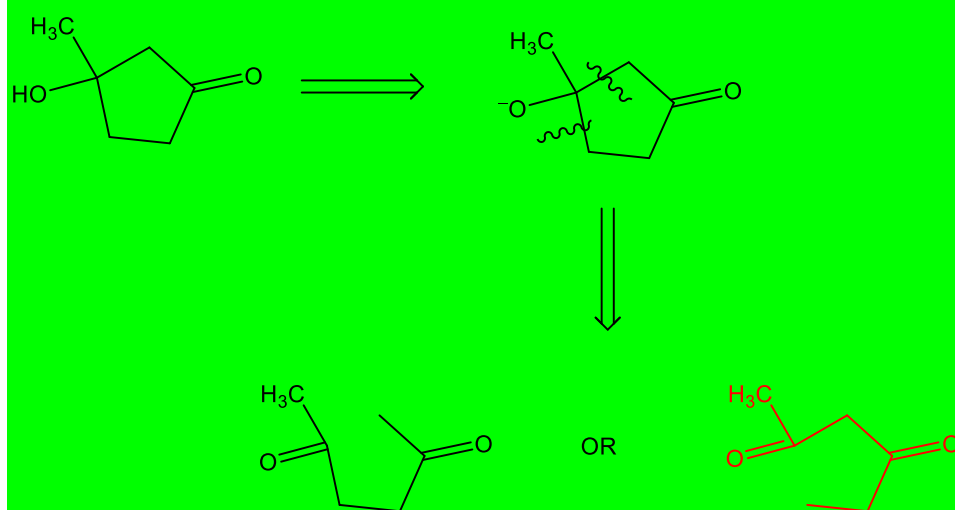


(ii)



Teachers' note:

Retrosynthesis



Structure in red rejected because only the hydrogen on a carbon adjacent to the carbonyl group can be removed. The structures are purposely left with the awkward bond angles so that it is easier to see the site of bond cleavage.

[Total: 20]